

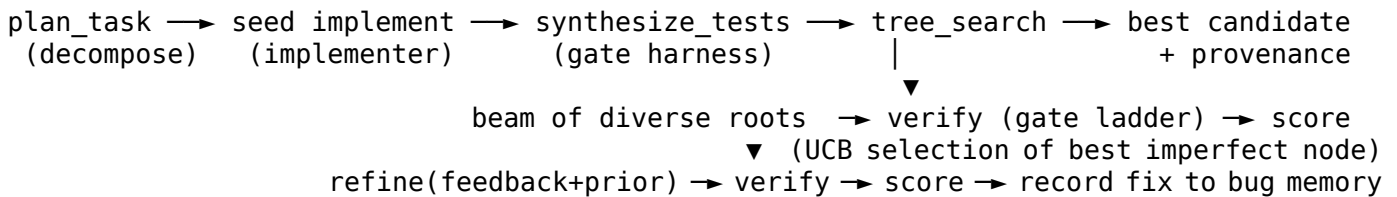
ELI Coding Agent (eli/coding/)

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A self-contained, **additive** subsystem that lifts ELI's code generation, analysis, and repair to frontier-grade. It touches nothing that already works: it lives in its own package and is reached through a new CODE_SOLVE action. The LLM is an *injected* generate callable (default = the local inference broker), so the whole pipeline is model-agnostic and unit-testable without a model.

The loop



Components

File	Capability
sandbox.py	Mandatory execution-feedback substrate. Runs code in a bounded, isolated subprocess (temp cwd, scrubbed env, MPLBACKEND=Agg, wall-clock timeout, POSIX RLIMIT_CPU; no RLIMIT_AS — it breaks numpy). Multi-language via a runner map (Python deepest). Crash detection is precise: only a genuine Python traceback counts; timeouts / signal kills / missing optional deps are tolerated so legitimate long/heavy/optional code isn't falsely failed. smoke_import verifies a patched module still loads.

File	Capability
bug_memory.py	Semantic bug classification + long-term memory. classify_bug maps a failure to a BugClass (null-handling, type, index/off-by-one, key-missing, name-undefined, import, zero-division, value, assertion, recursion, infinite-loop, logic-inversion, state-corruption, concurrency, resource-leak, io) with a stable, fuzzy-matchable <i>signature</i> (exception type + last frame + normalized message). BugMemory is a local SQLite store of (signature → class → fix) with exact-then-fuzzy (difflib) recall, so the agent learns from past fixes.
verification.py	Explicit verification gating + scoring + test synthesis. verify_candidate runs the ladder syntax → execution → tests, stopping at the first failure and attaching a BugDiagnosis + repair feedback. score_candidate aggregates a 0..1 score (tests dominate when present, else clean execution). synthesize_tests produces an executable harness (LLM-driven; deterministic smoke-harness fallback) that prints ELI_TESTS: <passed>/<total>.
planner.py	Structured decomposition — planner/implementer separation. plan_task → an approach + ordered steps (JSON, with a single-step fallback). implement writes/refines code against the plan; in refine mode it does patch-based incremental refinement (fix the specific feedback, keep prior code).
search.py	Tree search over solutions (not single-shot). A diverse beam of roots (temperature ladder = exploration), then iterative expansion choosing the node to refine by UCB1 ($\text{score} + c \cdot \sqrt{(\ln N / n_i)}$) — exploitation vs exploration. Refinements pull matching prior fixes from bug memory into the feedback; successful repairs are written back as (bug → unified-diff fix) pairs. Stops on target_score or budget.
agent.py	CodeAgent / solve() — composes all of the above; default broker-backed generator; returns best code + full provenance (plan, search trace, score, bug class).

Capability checklist (as requested)

- structured decomposition (planner/implementer separation) — planner.py ✓
- execution feedback loop (mandatory) — sandbox.py + verify_candidate gate 2 ✓
- tree search over solutions (not single shot) — search.py beam + UCB ✓
- explicit verification gating — verification.py gate ladder ✓
- long-term memory of bugs and fixes — bug_memory.py ✓
- automated test synthesis — verification.synthesize_tests ✓
- patch-based incremental refinement — planner.implement refine mode + search expansion ✓
- semantic bug classification (null handling, logic inversion, state corruption, ...) — bug_memory.classify_bug ✓
- scoring / “U-style” exploration / tree-of-code-solutions — score_candidate + UCB beam tree ✓
- multiple languages — sandbox runner map (Python deep; bash/js/ts/ruby/go/lua structural) ✓

Wiring & control

- New executor action **CODE_SOLVE** (args.description, optional args.language) saves the solution to artifacts/scripts/ and returns code + provenance. It is **additive** — GENERATE_SCRIPT and the self-patch engine are untouched.
- Env knobs: ELI_CODING_SANDBOX (default on; 0 disables execution), ELI_CODING_RUN_TIMEOUT (20s), ELI_CODING_BEAM (3), ELI_CODING_MAX_ITERS (6).
- Bug memory DB: <db_dir>/coding_memory.sqlite3.

Honest scope (what's verified vs model-dependent)

- **Verified deterministically** (tests in tests/test_coding_agent.py, no model): the sandbox crash/tolerate logic, the bug classifier + memory recall, the gate ladder + scoring, and the full plan→search→repair→record loop driven by a stub generator (buggy seed → UCB refinement → correct, fix recorded).
- **Model-dependent quality:** the *content* of plans, implementations, and synthesized tests scales with the local model. The machinery is frontier-shaped regardless of model.
- **Depth:** execution-level verification is Python-first; other languages run if their interpreter is installed (structural support).
- **Subtask DAG (added):** decomposable tasks are solved as a dependency graph of subtasks on the shared eli/core/dag engine (plan_graph.solve_dag), composed into one module — see dag.md. Gated by ELI_CODING_DAG (default on); cohesive tasks fall back to single-shot.
- **Chat routing (added):** CODE_SOLVE is now reachable from chat for explicit intents (“solve this coding problem”, “implement X and test it”, “use the coding agent”, “verified solution”). Generic “write a python function” still routes to the lighter GENERATE_SCRIPT (an earlier router path claims it — precedence is tunable but the router was intentionally left unreordered).
- **Not yet:** composition is single-module (ordered concat + import-dedupe), not true multi-file projects; test synthesis verifies against synthesized tests, not a user-supplied spec suite.